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2.3:

SKIN INTEGRITY & WOUND HEALING

WOUND

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- **Wound** is a disruption in the integrity of body tissue which may be intentional or unintentional
- Wound is classified depending on different categories.

Wound Classification

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- **Intentional wounds** occur during treatment or usually made under aseptic conditions, e.g. surgical incisions
- **Unintentional wounds** are unanticipated and are often the result of trauma or an accident.
- **Clean wounds** are intentional wounds that were created under conditions in which no inflammation was encountered
- **Clean-contaminated wounds are intentional wounds** that were created by entry into the alimentary, respiratory, genitourinary, or oropharyngeal tract under controlled conditions.

Wound Classification...

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- **Contaminated wounds** are open, traumatic wounds or intentional wounds in which there was a major break in aseptic technique, spillage from the gastrointestinal tract, or incision into infected urinary or biliary tracts.
- **Dirty and infected wounds** are traumatic wounds with retained dead tissue or intentional wounds created in situations where purulent drainage was present.

□ **Wound healing involves three phases:**

- Defensive(inflammatory) phase
- Reconstructive(proliferative) phase
- Maturation phase

1. Defensive (Inflammatory) Phase

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- Occurs immediately after injury and lasts about 3 to 4 days
- The major events that occur in this phase are **hemostasis and inflammation**
- **Hemostasis**, or cessation of bleeding, occurs by vasoconstriction of large blood vessels in the affected area.

Inflammatory Phase....

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- ***Formation of platelet plug, fibrin and fibrinous meshwork;***
 - Helps wound closure
 - Prevents excessive loss of blood and body fluid
 - Inhibits contamination of wound by m/osms

Inflammatory Phase cont...

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- **Inflammation** is the body's defensive adaptation to tissue injury and involves both vascular and cellular responses.
- Dilation of vessels by chemical mediators , results in increased blood flow and leakage of serous fluid to the surrounding tissues.
- During the cellular response, leukocytes move out of the blood vessel into the interstitial space.

2. *Reconstructive (Proliferative) Phase*

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- Begins on the third or fourth day after injury and lasts for 2 to 3 weeks.
- It involves:
 - Collagen deposition
 - **Angiogenesis** (formation of new blood vessels) begins within hours after the injury
 - Granulation tissue development and
 - Wound contraction.

3. *Maturation Phase*

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- Begins about the twenty-first day and may continue for up to 2 years or more, depending on the depth and extent of the wound.
- During this phase, the scar tissue is remodeled (reshaped or reconstructed by collagen deposition and lysis and debridement of wound edges).
- Although the scar tissue continues to gain strength, it remains weaker than the tissue it replaces.

Factors Affecting Wound Healing

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- **Age:** blood circulation, rate of cell growth and epithelilization and oxygen delivery to the wound may be impaired in the very young and the elderly
- **Nutrition:** balanced diet with adequate amounts of protein, carbohydrates, fats, vitamins, and minerals is needed
- **Drug therapy:** Steroids reduce inflammatory response and slow collagen synthesis.

Factors cont...

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- **Smoking:** decreases functional hemoglobin levels, impairing oxygenation to tissues.
- **Diabetes mellitus:** Small-vessel disease (microvascular changes) can impair tissue perfusion and oxygen delivery.
 - The high-glucose environment is an excellent medium for the growth of bacterial, fungal, and yeast infections.

Factors cont'd...

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- **Infection:** slows healing by prolonging the inflammatory phase of healing, competing for nutrients, and producing chemicals and enzymes that are damaging to the tissues.
- ***Dehiscence and Evisceration:***
 - **Dehiscence:** the partial or complete separation of the wound edges and the layers below the skin
 - **Evisceration occurs when the client's viscera protrude through the disrupted wound.**

Wound care

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1.Wound dressing

- ✓ Is process of covering wound or applying sterile protective covering using aseptic technique.

□ Types of wound dressing

- ✓ Clean wound dressing
- ✓ Septic wound dressing
- ✓ Wound dressing with drainage tube

Purpose of wound dressing

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- To Keep the wound moist and therefore enhance epithelization
- To Keep the wound clean
- To keep locally applied drugs in position
- To keep edges of the wound together by immobilization.
- To relief pain and comfort the patient

Purpose of wound dressing cont...

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- ☐ Protect wound from further injury
- ☐ Protect skin surrounding wound
- ☐ To absorb drainage
- ☐ To prevent hemorrhage.
- ☐ Provide physical, psychological, and aesthetic comfort
- ☐ Remove necrotic tissue
- ☐ Prevent, eliminate, or control infection

General Precautions in wound dressing

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- Wash hands thoroughly before, after and between several dressings.
- Dressing should be done after the wards have been cleaned
- Clean wounds should be dressed before wounds with drainage.
- Never do a dressing when the air is dusty from sweeping, bed making or in dust.
- Wear gloves when touching blood, body fluids, mucous membrane and handling soiled items with blood or body fluids.

General Precautions in wound dressing cont...

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- Do talk , cough and sneeze over wound
- Wear mask and protective eye wear as necessary
- Practice strict aseptic techniques to reduce transmission of micro organism
- Soaked wound with frequent drainage should be changed frequently.
- Clean wound from the cleanest area to the less clean area
- Separate instruments should be used for each dressing

General Precautions in wound dressing cont...

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- Keep forceps lower than the handles at all times
- Saline should be used to remove adherent dressings.
- Medicines are given and applied if necessary.
- Be economical on using gauze , sponges' applicators, adhesive or medication.
- Adhesive marks on skin can be removed with solvent such as ether alcohol or benzene.

1. DRESSING CLEAN WOUND

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- Clean wound dressing is aseptic technique of covering clean wound after cleaning .
- **Purpose**
 - To keep wound clean
 - To prevent the wound from injury & contamination
 - To keep in position drugs applied locally
 - To keep edges of the wound together by immobilization
 - To enhance wound healing
 - To relief pain and comfort the patient
 - To apply pressure

Equipment:

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Sterile dressing set

- One kidney dish
 - Sterile gloves
 - Cotton balls in a galipot
 - Sterile gauze (4×4 inch) or squares
- Sterile Dressing forceps (3)
Sterile Scissor
Sterile galipot
Sterile fenestrated towel (drape)
Spatula if ointment

Equipment Cont'd...

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Clean try

- Clean glove
 - Cleaning solution (Normal Saline, Sterile 0.9% sodium chloride), chlorhexidine, povidone-iodine, and hydrogen peroxide
 - Adhesive tape (Plaster)
 - Bath Blanket: (if needed)
 - Rubber and draw sheet
 - Waste Receiver(disposable plastic container)
- Bandage scissors or surgical blade
- Anti microbial Ointment: if prescribed
- Bath Blanket: (if needed)
- Screen
- Adhesive remover
- Protective apron: as the condition of the wound
- Chart

Procedure

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1. Check order for dressing change
2. Explain the procedure to the patient
3. Hand washing
4. Assemble the supplies at a convenient work area
5. Apply screen, close door and curtain.
6. Assist the patient to a comfortable position to expose the wound.
7. Place a rubber sheet under the patient to prevent soiling the linen.
8. Place opened, cuffed plastic bag near working area.
9. Loosen tape on dressing. Use adhesive remover if necessarily. If tape is soiled don gloves
10. Wear a protective apron when caring for a patient with a draining wound. Don non sterile gloves.

11. Gently remove and discard the old tape and soiled dressing in a plastic trash bag.

- ✓ Roll or lift an edge of the dressing, then gently remove it while supporting the surrounding skin. When possible, remove the dressing in the direction of hair growth.
- ✓ If the dressing sticks to the wound, moisten with sterile normal saline and then remove.
- ✓ Sterile saline provides for easier removal of dressing. Assess amount, type and odor of draining if present

12. Remove and discard non sterile gloves.

13. Using aseptic technique open the packed sterile instruments, sterile dressings, the irrigation and cleaning solution, and the instrument set to provide a sterile field.

Procedure cont...

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14. Pour cleaning solution to galipot, gauze and cotton from a drum.
15. Don sterile gloves
16. Gently remove and discard the old tape and soiled dressing in a plastic trash bag. Roll or lift an edge of the dressing, then gently remove it while supporting the surrounding skin. When possible, remove the dressing in the direction of hair growth. If the dressing sticks to the wound, moisten with sterile normal saline and then remove. Sterile saline provides for easier removal of dressing. Assess amount, type and odor of draining if present
17. Remove and discard the soiled gloves.

Procedure cont...

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18. Using aseptic technique open the packed sterile instruments, sterile dressings, the irrigation and cleaning solution, and the instrument set to provide a sterile field, Pour cleaning solution to galipot, gauze and cotton from a drum.
19. Don sterile gloves.
20. Apply fenestrated towel to the wound to increase the sterile field
21. If sample is needed, take the sample first then clean.
22. Take the second sterile forceps, and clean wound with cotton balls soaked in antiseptic solution starting from inside to the outside.

Procedure cont...

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23. Use one gauze square for each wipe, discard each square by dropping in to plastic bag, do not touch bag with forceps.
24. Again use the second forceps to dry wound using gauze sponge and same motion by another new forceps then discard.
25. Apply medication if any and dress the wound with sterile gauze with sterile another dressing forceps
 - ✓ Ointment and paste must be smeared with spatula on gauze and then applied on the wound.
 - ✓ Solutions or powder can be applied direct on the wound.
26. Make sure that the wound is properly covered

Procedure cont...

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27. Fix dressing in place using adhesive tape or bandage.
28. Remove fenestrated towel, rubber and draw sheet.
29. Remove gloves from inside out, and discard them in plastic waste bag.
30. Provide patient comfort measures.
31. Clean and return equipment to proper place.
32. Wash your hands
33. Document the procedure

2. Dressing septic wound

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- **Dressing septic wound:** is method of covering infected wound that generally contain purulent material (pus) draining from the wound site.
- **Purposes:**
 - Absorb materials being discharged from the wound.
 - Apply pressure to the area.
 - Apply local medication.
 - Prevent pain, swelling & injury.

Dressing septic wound cont...

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□ Precautions

- If sterile forceps are not available use sterile gloves.
- Immerse used forceps, scissors and other instrument in strong antiseptic solution before cleansing and discard soiled dressing properly
- In a big ward it is best to give priorities to clean wounds and then to septic wounds when changing dressings as this might lessen the risk of cross infection
- Wounds should not be too tightly packed in effort to absorb discharge as this may delay healing.

Equipment

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Sterile Dressing set

- Sterile galipot
 - Sterile kidney dish
 - Sterile gauze
- Sterile forceps (3) Sterile test tube or slide if necessary
- Sterile cotton tipped application
- Sterile pair of gloves if needed in case of gas gangrene rabies etc

Equipment cont...

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Clean try

- Clean glove
 - Surgical glove
 - Cleaning solution (normal saline, H₂O₂)
 - Ordered medication
 - Plaster
 - Bandage scissors or surgical blade
 - Probe and director if required
 - Benzene or ether
 - Bandage or adhesive tape and
- Bucket to put in soiled dressing/water proof disposable bag
- Rubber and cotton draw sheet /Mackintosh with its cover
- Spatula if ointment
- Receiver with strong disinfectant to immerse used instrument
- Bucket to put in soiled dressing

Procedure

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1. Check order for changing dressings/dressing
2. Great the patient and explain procedure to the patient.
3. Wash your hands
4. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
5. Put on single use examination glove

Procedure cont...

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6. Apply screen (close door and curtain), drape, & put patient in comfortable position
7. Place rubber sheet and its cover under the patient to prevent soiling the linen.
8. Place a clean towel or draw sheet underneath the working area to minimize contamination.
9. First remove the outer layer of the dressing.
10. Remove the outer layer of dressing and dispose with glove
11. Wear sterile gloves

Procedure cont...

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12. Use forceps to remove the inner layer of the dressing smoothly & discard forceps
13. Observe wound and check if there is drainage rubber or tube.
14. Take specimen for culture or slide if ordered (do not cleanse wound with antiseptic before you obtain the specimen).
15. Take the sterile cleaning forceps
16. Start cleaning wound from the cleanest part of the wound to the most contaminated part using antiseptic solution (Hydrogen peroxide 3% is commonly used for septic wound).

Procedure cont...

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17. Debride dead tissue as needed
18. Discard cotton ball used for cleaning after each stroke over the wound.
19. Cleanse the skin around the wound to remove the plaster gum with benzene
20. Use gauze for drying the skin around properly.
21. Use third forceps for dressing the wound
22. Dress the wound and make sure that the wound is covered completely.
23. Make sure that the wound is properly covered

Procedure cont...

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24. Fix dressing in place with adhesive tape or bandages.
25. Remove gloves from inside out, and discard them in plastic waste bag.
26. Leave patient comfortable and tidy.
27. Clean and return equipment to proper place.
28. Wash your hands
29. Document the procedure
30. Send the specimen to laboratory

3. Dressing with drainage tube

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- Dressing with drainage tube is the method of flushing wound with plenty of sterile fluid
- **Purpose**
 - ❖ Aids to prevent hematoma or collection of fluid in the affected area

Dressing with drainage tube cont...

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□ **Precautions**

- Safe method should be used for disposing old dressing.
- Gauze and cotton used for cleaning wound.
- Take preventive measure to avoid skin irritation and excoriation.
- If drainage tube is attached to the bottle precaution must be taken to secure the tube in place and avoid the risk of cross infection

Dressing with drainage tube cont...

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Equipment

- ❑ **Sterile Field Set**
 - ❑ Sterile kidney dish
 - ❑ Sterile galipot
 - ❑ Sterile scissors
 - ❑ Sterile forceps(3)
 - ❑ Sterile cotton balls
 - ❑ Sterile gauze
 - ❑ Sterile safety pins if needed

☐ **Clean tray**

- ☐ Antiseptic solution as ordered
- ☐ Cotton wool or absorbent
- ☐ Receiver
- ☐ Rubber sheet and its cover
- ☐ Adhesive tape or bandage
- ☐ Dressing scissors
- ☐ Ointment paste or paraffin gauze
- ☐ Spatula if needed
- ☐ One pair sterile gloves if available

Dressing with drainage tube cont...

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Procedure

1. Check order for dressing / changing dressing
2. Great the patient
3. Check the order and the site of dressing drainage tube
4. Explain procedure to the patient.
5. Perform hand hygiene
6. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
7. Put on single use examination glove

Dressing with drainage tube cont...

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8. Apply screen or close door and curtain
9. Drape and position the patient according to the need and put rubber sheet and its cover under the part to be dressed.
10. Remove the outer layer of the dressing.
11. Use sterile forceps and remove the inner layer of the dressing (pay attention so that the drainage tube is not pulled out with the old dressing).

Dressing with drainage tube cont...

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12. Observe the wound for the type and amount of discharge.
13. Clean the wound with cotton balls soaked in antiseptic solution.
14. Grasp the top of drainage tube with sterile forceps. Pull it up a short distance while using gentle rotation and cut off the tip of the drain with sterile scissors (the length to be cut depends on the instruction or order).
15. Place sterile safety pin through the drainage tube close to the wound using sterile gloves or sterile gauze, if it is in the abdomen to stop the drainage tube slipping down out of sight.

Dressing with drainage tube cont...

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16. Make sure the wound and the skin around are properly cleaned.
17. Apply ointment or paste to the skin with spatula directly around to prevent irritation and excoriation (if the excoriation exist use paraffin gauze to prevent further complications).
18. Cut the gauze towards its center to fit around rubber drainage tube so that it fits properly around the tube thus preventing discomfort.
19. Use adhesive tape or bandage to secure the dressing in place.
20. Record state of wound and the drainage.

4. Wound irrigation

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- **Wound irrigation** is the process of washing debris, drainage, or exudates out of the wound to promote healing.
- **Purpose**
 - To clean the area from pathogens and debris
 - To apply local heat
 - To irrigate with antiseptic solutions
 - To promote wound healing from the deepest area of a wound to the skin surface.
 - To remove and monitor drainage from wounds.
 - To manage wounds while preventing infection

□ **Precaution**

- Keep patient in convenient position.
- According to the need the solution will flow from wound down to the receiver.
- Use sterile technique and warm solution for irrigating the wound.

Equipment

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Sterile Field Set

- ☐ Sterile gauze
- ☐ Sterile gloves
- ☐ Sterile forceps (3)
- ☐ Sterile test tube or slide
- ☐ Sterile cotton tipped application
- ☐ Sterile galipot or kidney dish
- ☐ Sterile catheter
- ☐ Sterile syringe 20 cc
- ☐ Gown
- ☐ Goggles
- ☐ Mask

Wound irrigation cont...

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Clean tray

- Clean glove
- Bandage scissors or surgical blade
- Bandage or adhesive tape and Bucket to put in soiled dressing
- Receiver (2)
- Rubber sheet and its cover
- Solution (hydrogen peroxide or normal saline commonly used)
- Adhesive tape or bandage
- Bandage scissors
- Receiver for soiled dressing

Procedure

1. Check order for irrigation
2. Great the patient
3. Check the order for irrigation and dressing
4. Explain procedure to the patient.
5. Wash your hands
6. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
7. Put on single use examination glove

Wound irrigation cont...

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8. Apply screen (close door and curtain) and drape
9. Position patient comfortably to permit gravitational flow of irrigation solution wound and into collection receptacle
10. Put rubber sheet and its cover under the part to be irrigated.
11. Check temperature of solution
12. Apply gown and goggles if needed
13. Remove the outer layer of the dressing by disposable glove and dispose the glove

Wound irrigation cont...

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14. Open sterile field
15. Put on sterile glove
16. Remove the inner layer of the dressing using the first sterile forceps.
17. Put the sterile receiver under patient to receive the out flow(may be sterile basin)
18. Use syringe with desired amount of solution fitted with the catheter.
19. Use forceps to direct the catheter into the wound.

Wound irrigation cont...

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20. First inject the solution such as hydrogen per oxide at body temperature gently and wait for the flow. This must be followed by normal saline for rinsing.
21. Make sure the wound is cleaned and dried properly with gauze.
22. Dress the wound and check if it is covered completely.
23. Secure dressing in place with adhesive tape or bandage.
24. Leave patient comfortable and tidy.
25. Clean and return equipment to proper place.
26. Wash your hands
27. Record the state of the wound.

Suturing

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- **Suturing** is the technique of uniting parts of the body by stitching them together.
- **Sutures** are threads used to sew body tissue together which can be **absorbable** (**Chromic Cat gut**) and **non absorbable** (silk, cotton, linen, clips and wire nylon).
- It is also synonymous with ***stitch***.
- Absorption takes place in from 3 days to 3 months for absorbable one.

Types of Suture Materials

The most common absorbable sutures

A) Surgical Catgut:

- ☐ *Collagen manufactured from the submucosa of sheep intestine or serosa of beef intestine.*
- ☐ *Used in tissue that heals rapidly.*
- ☐ *Digested by body enzymes and absorbed by tissue so that no permanent foreign body remains.*

❖ Plain Surgical Catgut

- ✓ Lose tensile strength quickly, usually in 5 to 10 days
- ✓ Used to ligate small vessels and to suture subcutaneous fat
- ✓ Available in sizes **3 to 6-0**.
- ✓ These sutures are used only externally on skin, not internally, particularly for facial cosmetic surgery.

❖ **Chromic Surgical Catgut**

- ✓ It is used for ligation of larger vessels and for suture of tissues in which nonabsorbable suture materials are not recommended
- ✓ Available in sizes **3 to 7-0**

B. Synthetic Absorbable Sutures

- Are absorbed by a slow hydrolysis process in the presence of tissue fluids.
- They are used for ligating and suturing.
- Can be used in nearly all tissues.
- ❖ The sutures included in this group are:
 - Polyglycolic acid (Dexon) suture
 - Polyglactin 910 (Vicryl) suture
 - Polydioxanone (PDS) Suture
 - Polyglycaprone (Monocryl) suture

Non absorbable Suture

- It is used in tissues that heal more slowly than those requiring any type of absorbable suture.

A. Surgical Silk

- An animal product made from the fiber spun by silkworm larvae in making their cocoons
- Widely used non-absorbable suture that is easy to handle and is both supple and strong.

Non absorbable Suture cont...

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B. Surgical Cotton

- ✓ Manufactured from the fibers of the cotton plant
- ✓ Supple and easy to handle
- ✓ Has inferior strength and tendency to flake
- ✓ Can be strengthening by dipping it into saline solution prior to use.
- ✓ Its application is nearly identical to that of silk.

C. Polyester Suture

- ✓ The strongest of all sutures except for surgical steel
- ✓ It is usually multifilamented and may be coated with Teflon
- ✓ Used in a wide variety of tissues, including facial, cardiovascular, and ophthalmic.

Non absorbable Suture cont...

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D. Nylon Suture

- ✓ Used primarily for skin closure, ophthalmic procedures, and microsurgery.
- ✓ Produces minimal tissue reaction
- ✓ Has high tensile strength, and resists capillary action
- ✓ The major disadvantages of nylon are its elasticity and stiffness, which necessitate the laying of many knots.

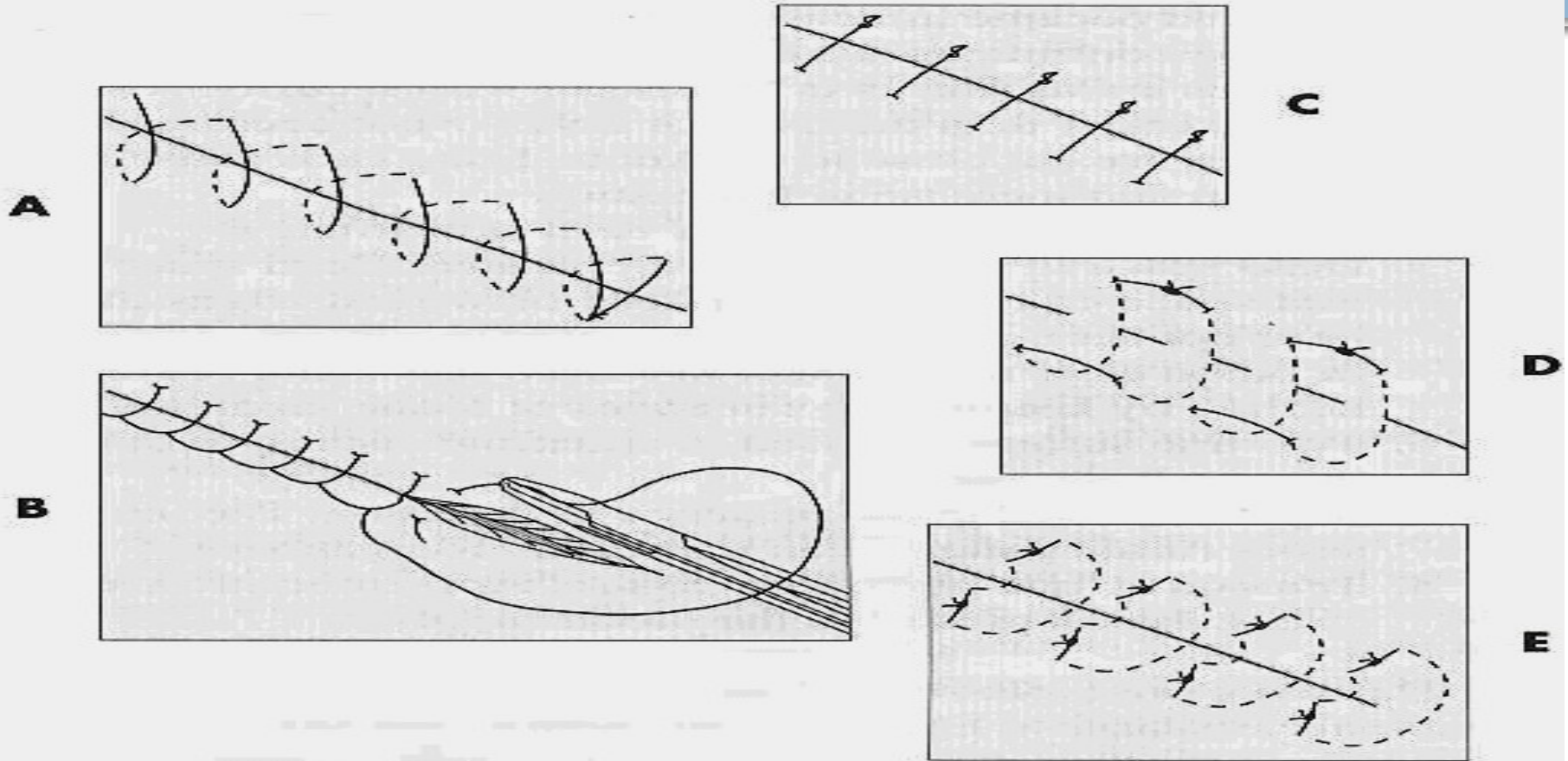
Factors that influence the choice of suture materials

- ✓ Biologic characteristics of the suture material
- ✓ Healing characteristics of the tissue
- ✓ Location and length of the incision
- ✓ Presence or absence of contamination and/or infection
- ✓ Patient problems such as obesity, debility, advanced age and diseases
- ✓ Physical characteristics of the material such as ease of passing through tissue, knot tying and other personal preference of the surgeon.

Methods of Suturing

- There are two basic methods of suturing and various ways to utilize the two techniques.
- The suture is either ***running (continuous)*** or else it is ***interrupted***.

Methods of Suturing cont...



**A and B: continuous;
C, D, and E: interrupted.**

Purpose

- To approximate wound edges until healing occurs.
- To speed up healing of wound.
- To minimize the chance of infection
- For aesthetic purpose

Indication

- Open intentional and unintentional wound

Contraindication

- Edema of the wound margins
- Infection
- Puncture wounds
- Animal bites
- Tendon, nerve, or vessel involvement
- Wound more than 12 hours old (body) and 24 hrs (face)

Precautions

- ❑ Check that the patient gets TAT before he leaves the hospital.
- ❑ Do not suture puncture (deep) wound.
- ❑ Before you suture any wounds make sure it is free of any foreign body.
- ❑ The completed knot must be tight, firm, and tied

Suturing cont...

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- To avoid wicking of bacteria, knot should not be placed in incision lines
- Knots should be small and the ends cut short (2-3mm)
- Avoid excessive tension to finer gauge materials as breakage may occur
- Avoid using a jerking motion, which may break the suture
- Do not tie suture too tightly as tissue ischemia may occur.

Equipments

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Sterile field set

- Tray or trolley covered with a sterile towel
- Sterile needle holder
- Sterile suture scissors
- Sterile round needle (2)
- Sterile cutting needle (2)
- Sterile hole towel (fenestrated towel)
- Sterile kidney dish(2)
- Sterile needle and syringe
- Sterile gloves
- Sterile tissue forceps(2)
- Sterile cotton swabs in gallipots
- Sterile gauze
- Sterile artery forceps(3)
- Sterile needle holder (2)

□ **Clean tray**

- Silk and catgut
- Antiseptic solution with its container
- Adhesive plaster
- Local anesthesia
- Light bulb

Procedure

1. Check the order for suturing
2. Great the patient, introduce yourself and explain the purpose of the procedure to the patient.
3. Wash your hands
4. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
5. Adjust light
6. Put on sterile gloves.

Suturing cont...

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7. Clean the wound thoroughly.
8. Drape the wound with the hold sheet
9. Infiltrate the edge of the wound to be sutured with local anesthesia.
10. Approximate the edge of fascia with the help of the tissue forceps and using the round needle and catgut.
11. Suture the facial layer first followed by muscles. Suture the facial layer and muscle with chromic catgut and.

Suturing cont...

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12. Using cutting needle and silk suture the outer layer of skin approximating the edges with the help of the tissue forceps
13. Clean wound area with iodine
14. Dress the wound with sterile gauze.
15. Remove the whole sheet.
16. Make the patient comfortable.
17. Remove all equipment wash and return to its proper place or send for sterilization.
18. Wash your hands
19. Record the state of the wound.

Stitch removal

- Stitch removal is the process of removing suture threads
- **Purpose**
 - To prevent further contamination
 - To comfort the patient
 - To relieve tension from wound site.

Precaution

- Keep wound clean and dry for the first 24 hours.
- Bathing is allowed after 48 hours.
- Stick to the day stitches should be removed.
- Keep wound clean and make dry dressing if no discharge.
- Change bandages often.

Equipment

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Sterile field set

- Sterile gauze
- Sterile cotton balls
- Sterile kidney dish
- Sterile forceps (2)
- Toothed tissue /pick up/pin forceps (1)
- Sterile stitch scissors/surgical blade

Clean tray

- Rubber sheet and its cover
- Antiseptic solution
- Receiver/ waste container
- Adhesive tape/ bandage
- Plaster scissor

Procedure

1. Check the order for stitch removal
2. Greet the patient and Explain the purpose of procedure.
3. Wash your hands
4. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
5. Adjust light
6. Position the patient and protect bedding with rubber sheet and its cover.
7. Wear sterile glove

Stitch removal cont...

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8. Remove the old dressing and discard it in the receiver.
9. Cleanse the wound using the 1st forceps with antiseptic solution starting from the cleanest part of wound to the most contaminated part.
10. Place sterile gauze to receiver pieces of sutures.
11. Take a scissor in the right hand/ surgical blade.
12. Take pick up/pin set/ toothed tissue forceps in the left hand.
13. Pull up gently the knot resting against the skin with the forceps pass the point of the scissors under the knot then cut the stitch on one side and remove.

Stitch removal cont...

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14. Receiver pieces of stitches on sterile gauze and count number of stitches removed.
15. Inspect the wound for healing and apply iodine on suture site.
16. Apply dressing accordingly.
17. Keep patient comfortable and tidy.
18. Perform hand hygiene
19. Clean and return equipment to their proper place
20. Record the state of the wound.

Clips Application

- **Clips** are metal sutures used to stitch the skin

Purpose

- To approximate wound edges until healing occurs
- To promote wound healing
- To minimize the chance of infection
- For aesthetic (cosmetics) purpose

Equipment

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- Tray or trolley covered with a sterile towel
- Sterile hole towel (fenestrated towel)
- Sterile kidney dish
- Sterile needle and syringe
- Sterile gloves
- Mitchell's clip applier
- Tissue forceps (Tooth dissecting forceps)
- Sets of clip
- Sterile cotton swabs in gallipots
- Sterile Gauze

□ **Clean tray**

- Antiseptic solution with its container
- Transferring with its container
- Adhesive plaster
- Local anesthesia (Lidocaine)

Procedure

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1. Check the order for clips application
2. Great the patient and explain the purpose of clips application to the patient.
3. Wash your hands
4. Wear clean glove
5. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
6. Adjust light
7. Wash your hand again
8. Wear sterile glove
9. Clean the wound thoroughly

Cont...

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10. Drape the wound with the hold sheet
11. Infiltrate the edge of the wound to be sutured with local anesthesia.
12. Approximate the edge of skin with the help of the tissue forceps and apply clip to the wound edge using the Mitchell's clip applier.
13. Clean the wound and cover it with sterile gauze.
14. Remove the fenestrated/window towel .
15. Make the patient comfortable.
16. Remove all equipment wash and return to its proper place or send for sterilization.
17. Wash your hands
18. Record the state of the wound.

Removal of clips

- **Removal of clips** is removal of clips from sutured area.

Purpose

- To prevent further contamination.

Precaution

- Stick to the day clips should be **removed**.
- Use aseptic technique
- Keep wound clean and make dry dressing if no discharge.
- Change bandages often.

Equipment

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Sterile field set

- ☐ Sterile gauze
- ☐ Sterile cotton balls
- ☐ Sterile kidney dish
- ☐ Sterile forceps (3)
- ☐ Sterile clip removal forceps

Clean tray

- ☐ Rubber sheet and its cover
- ☐ Receiver/ waste container
- ☐ Adhesive tape/ bandage
- ☐ Plaster scissor
- ☐ Antiseptic solution (Savalon 1% and iodine)
- ☐ Benzene or ether

Procedure

1. Check the order for clips application
2. Great the patient and explain the procedure.
3. Wash your hands
4. Wear clean glove
5. Clean trolley or tray, assemble sterile equipment on one side & clean items on the other side and make sure that the sterile equipments are properly covered.
6. Adjust light
7. Wash your hand again
8. Position and drape the patient.

Cont'd...

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9. Protect the bedding with rubber sheet and its cover.
10. Wear sterile glove
11. Remove old dressing and discard.
12. Cleanse wound with antiseptic solution starting from cleanest part of the wound to the most contaminated part and discard the cotton ball.
13. Place sterile gauze to receive removed clips.
14. Take clip remove with the right hand and dissecting forceps with the left hand.

Cont'd...

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15. Insert the lower blade of the clip remove below the middle of clip using the dissecting forceps as a support of old the clips in place, and close the blade firmly as this will cause disagreement of the clips from the skin.
16. Receive clips on sterile gauze and count the number of clips.
17. Apply iodine on the skin puncture in place with adhesive tape.
18. Leave patient comfortable and tidy.
19. Record the state of scar.
20. Clean and return used equipment to its proper place.

THANK
YOU

